# Retail Demand Forecasting for Intermittent Demand in Consumer Packaged Goods

*A Comparative Study of Traditional and Specialized Forecasting Methods*

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# 1. Introduction

## Section Objective

This section introduces the business context of the project, explains the importance of demand forecasting in the Consumer Packaged Goods (CPG) industry, and presents the motivation for studying intermittent demand. It also outlines the objectives and scope of the forecasting framework developed during the internship.

Consumer Packaged Goods (CPG) are products that are purchased frequently by consumers and typically have a short consumption cycle. These include everyday items such as food products, beverages, personal care products, and household essentials. Due to their high sales volume and extensive distribution networks, efficient inventory planning is a critical requirement for organizations operating in the CPG sector.

Demand forecasting plays a central role in inventory and supply chain management by estimating future product demand using historical sales data. Accurate forecasts enable businesses to maintain optimal inventory levels, reduce stockouts, minimize excess inventory, improve customer service, and support data-driven operational planning. Conversely, inaccurate forecasts may lead to increased inventory costs, lost sales, and inefficient resource utilization.

While forecasting methods perform well for products with consistent sales patterns, many retail products exhibit **intermittent demand**, where sales occur irregularly and are separated by multiple periods of zero demand. Such demand patterns are common among low-velocity Stock Keeping Units (SKUs), seasonal products, spare parts, and niche retail items. These characteristics violate the assumptions made by many traditional forecasting techniques, making intermittent demand forecasting a specialized problem within time-series analysis.

The objective of this project was to analyse historical retail sales data, understand the characteristics of intermittent demand, and evaluate forecasting techniques suitable for sparse demand series. The study involved exploratory data analysis (EDA), demand characterization, granularity analysis, and the implementation of both traditional and intermittent demand forecasting methods. Particular emphasis was placed on understanding demand sparsity, selecting an appropriate forecasting granularity, and comparing forecasting models designed for intermittent demand scenarios.

The findings of this study provide insights into forecasting sparse retail demand while highlighting the importance of selecting forecasting techniques that align with the underlying demand behaviour. The resulting framework serves as a foundation for improving inventory planning and supporting more reliable demand forecasting in Consumer Packaged Goods (CPG) analytics.

# 2. Business Problem

## Section Objective

This section describes the business challenge addressed during the project and explains why accurate forecasting of intermittent demand is important for inventory planning and operational decision-making in the Consumer Packaged Goods (CPG) industry.

## 2.1 Business Context

Consumer Packaged Goods (CPG) companies manage thousands of Stock Keeping Units (SKUs) across multiple retail outlets. Maintaining appropriate inventory levels for each product is a continuous challenge because customer demand varies across products, stores, seasons, promotions, and external market conditions. While high-selling products generally exhibit stable demand patterns, a significant proportion of products experience irregular purchasing behaviour, making demand prediction considerably more difficult.

Products with infrequent or sporadic sales often generate long sequences of zero-demand periods followed by occasional non-zero demand. This demand pattern, commonly referred to as **intermittent demand**, is frequently observed in low-velocity products, specialty items, seasonal merchandise, and slow-moving inventory. Traditional forecasting techniques typically assume relatively continuous demand and therefore struggle to produce reliable forecasts for such products.

## 2.2 Problem Statement

Retail organizations must balance inventory availability against inventory holding costs across thousands of store-product combinations. While products with regular purchasing patterns can often be forecast using conventional time-series methods, many low-velocity SKUs exhibit intermittent demand characterized by frequent zero-sales periods and irregular purchasing behaviour.

These demand characteristics make traditional forecasting approaches less reliable, often resulting in either excess inventory or stockouts. Consequently, selecting an appropriate forecasting methodology for intermittent demand becomes an important business problem rather than simply a modelling exercise.

This project aims to identify an appropriate forecasting granularity, analyse the characteristics of intermittent retail demand, evaluate forecasting techniques designed for sparse demand, and determine the approach that provides the most reliable forecasting performance for store-product-week level demand.

## 2.3 Business Impact

Accurate forecasting of intermittent demand enables organizations to make informed inventory decisions by reducing stockouts, minimizing excess inventory, and improving product availability. Reliable forecasts also support more efficient replenishment planning, warehouse operations, and supply chain management. By selecting forecasting techniques that appropriately capture sparse demand behaviour, organizations can improve operational efficiency while reducing inventory-related costs associated with inaccurate demand estimation.

# 3. Project Objectives

## Section Objective

This section outlines the primary objectives of the project and defines the scope of the work carried out during the internship. These objectives guided the exploratory analysis, model development, and evaluation process presented in the subsequent sections.

## 3.1 Project Objectives

1. Analyse historical retail sales data to understand demand characteristics and identify the presence of intermittent demand patterns.
2. Determine an appropriate forecasting granularity by evaluating the impact of data aggregation on demand sparsity and forecasting stability.
3. Perform comprehensive Exploratory Data Analysis (EDA) to identify demand trends, promotional effects, seasonal behaviour, and data quality considerations that influence forecasting performance.
4. Implement and evaluate both traditional and intermittent demand forecasting techniques, including Moving Average, Croston's Method, Syntetos-Boylan Approximation (SBA), and Teunter-Syntetos-Babai (TSB).
5. Compare forecasting models using appropriate evaluation metrics to determine the most suitable approach for intermittent retail demand forecasting.
6. Derive business insights from the analysis and provide recommendations that support inventory planning and demand forecasting within a Consumer Packaged Goods (CPG) environment.

# 4. Dataset Description

## Section Objective

This section describes the dataset used in the project, including its structure, key variables, and characteristics. Understanding the available data is essential before performing preprocessing, exploratory analysis, and forecasting model development.

## 4.1 Dataset Overview

The dataset used in this project consists of transactional retail sales records collected from multiple stores within a Consumer Packaged Goods (CPG) environment. Each record represents the sales activity of a specific product on a given calendar date and includes information related to sales volume, pricing, promotional activity, and store-product identifiers. The dataset provides the historical demand required for analysing purchasing behaviour and developing forecasting models for intermittent demand.

The original dataset contains approximately **3.3 million daily sales records**, covering **200 retail stores** and **50 products**. This extensive coverage captures diverse demand patterns across different store-product combinations and provides sufficient historical observations for analysing sparsity, promotional influence, seasonality, and temporal demand behaviour. To support forecasting at an appropriate temporal resolution, the daily transactional data was later aggregated to weekly demand during the data preparation stage.

## 4.2 Dataset Variables

| Variable | Data Type | Description |
|---|---|---|
| **calendar_date** | Date | Date on which the sales transaction occurred. |
| **masked_store_id** | Integer | An anonymized identifier representing each retail store. |
| **masked_product_id** | Integer | An anonymized identifier representing each product (SKU). |
| **sales_value_ty** | Double | Total sales value generated by the product on that date. |
| **sales_volume_ty** | Double | Quantity of product sold on the transaction date. This serves as the primary demand variable. |

## 4.3 Data Characteristics

Several characteristics of the dataset make it suitable for studying intermittent demand forecasting. First, the data consists of daily retail transactions collected across multiple stores and products, providing substantial variation in demand behaviour. Second, the use of anonymized store and product identifiers preserves business confidentiality while maintaining the relationships necessary for analytical modelling. Finally, the availability of pricing and promotional information enables subsequent analysis of external factors influencing demand variability.

However, before meaningful forecasting could be performed, the daily transactional data required preprocessing and aggregation to a suitable temporal granularity.

# 5. Data Preparation

## Section Objective

This section describes the preprocessing steps performed to transform the raw transactional dataset into a structured weekly time-series dataset suitable for exploratory analysis and forecasting. The preparation process focused on reducing demand sparsity, establishing a consistent temporal structure, and generating variables required for subsequent analysis.

## 5.1 Weekly Aggregation

The original dataset contained daily sales transactions, which exhibited a high proportion of zero-demand observations across many store-product combinations. Such sparsity increases forecasting instability and makes it difficult to identify meaningful demand patterns at the daily level.

To address this challenge, daily transactions were aggregated into weekly observations. Weekly aggregation preserves underlying purchasing behaviour while reducing short-term fluctuations and excessive zero-demand periods. Compared with daily granularity, the weekly representation provides a more stable time series without excessively smoothing demand, making it an appropriate level of aggregation for intermittent demand forecasting.

## 5.2 Weekly Dataset

Following aggregation, each record represented the weekly demand of a specific product at an individual store. Weekly sales volume, total sales value, and average selling price were computed for every store-product-week combination. The resulting dataset contained approximately **478 thousand** weekly observations spanning **70 weeks**, providing a consistent structure for subsequent exploratory analysis and forecasting.

Table 2. Weekly Aggregated Dataset Variables

| Variable | Description |
|---|---|
| **week_id** | Sequential identifier assigned to each week for indexing and visualization purposes. |
| **week_start** | Calendar date representing the first day of the corresponding week. |
| **masked_store_id** | Anonymized store identifier. |
| **masked_product_id** | Anonymized product (SKU) identifier. |
| **weekly_volume** | Total sales volume recorded during the week. |
| **weekly_value** | Total sales value generated during the week. |
| **weekly_avg_price** | Average selling price of the product during the week. |

## 5.3 Week Identifier Generation

A sequential **week_id** was generated after ordering the data chronologically using **week_start**. The sequential identifier simplifies indexing, visualization, and aggregation throughout the analysis. However, the true temporal sequence of the dataset was always preserved using the **week_start** variable during analysis and forecasting to ensure chronological consistency.

## 5.4 Data Preparation Outcome

The data preparation process transformed raw daily transactions into a structured weekly demand dataset suitable for intermittent demand analysis. The resulting dataset reduced daily sparsity, maintained meaningful purchasing patterns, and established a consistent time-series structure for all store-product combinations. This prepared dataset served as the foundation for the exploratory data analysis presented in the following section.

# 6. Exploratory Data Analysis

## Section Objective

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the weekly demand data before developing forecasting models. Rather than serving as a descriptive exercise, the analysis aimed to identify demand behaviour, quantify intermittency, evaluate the effects of promotions and holidays, assess demand variability across stores and products, and determine an appropriate forecasting granularity. The insights obtained during this stage directly informed feature engineering, model selection, and the overall forecasting methodology adopted in this project.

## 6.1 Analytical Roadmap

The exploratory analysis was organised into a sequence of investigations, each designed to answer a specific business or modelling question. Rather than analysing individual variables in isolation, the EDA progressively examined demand characteristics, promotional behaviour, seasonal effects, product and store variability, and demand intermittency. Each stage provided evidence that supported subsequent modelling decisions and justified the forecasting framework adopted later in the project.

1. Weekly demand characteristics
2. Promotion analysis
3. Store-level demand analysis
4. Product-level demand analysis
5. Holiday and seasonal analysis
6. Demand intermittency classification
7. Key analytical findings

## 6.2 Promotion Identification and Detection

### Purpose

The original dataset did not contain an explicit indicator identifying promotional periods. Since promotional activity is a major driver of retail demand, it was necessary to infer promotional events before analysing their impact on sales behaviour. Promotion detection was therefore performed by estimating a regular selling price for each store–product combination and identifying temporary price reductions relative to this baseline. This approach enabled subsequent analyses of promotional effectiveness and allowed promotion-related features to be incorporated into the forecasting framework.

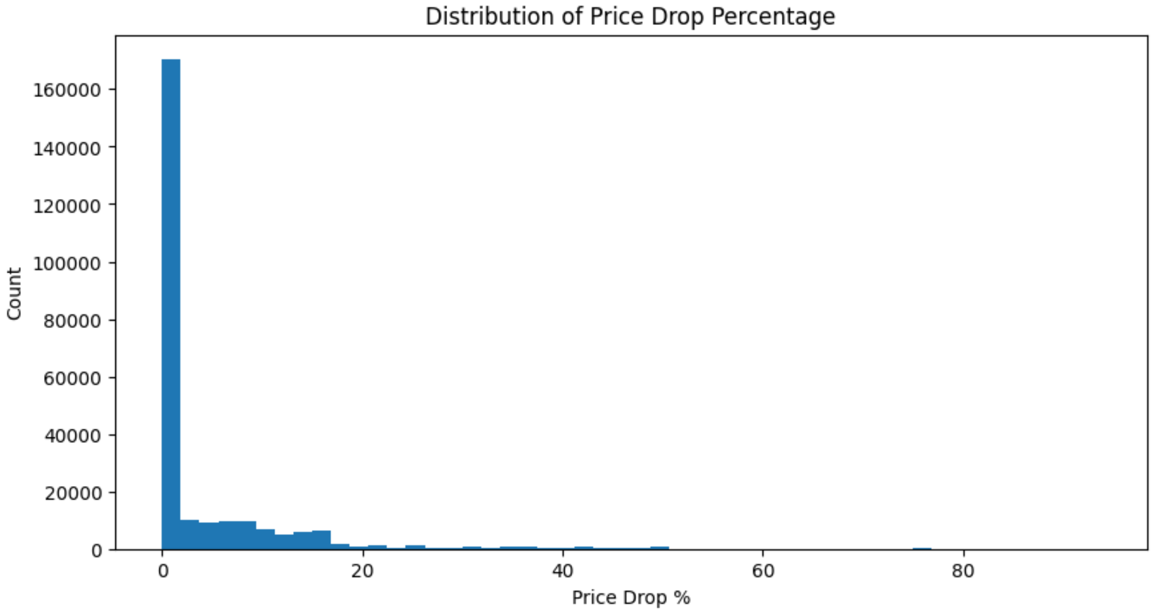


Figure X. Distribution of Price Drop Percentage Used for Promotion Detection

### Key Observation

The distribution of price reductions indicates that most observations experience little or no discount, while only a relatively small proportion of records exhibit substantial price decreases. Using the price-drop methodology, approximately **12% of store–product–week observations** were classified as promotional periods. This indicates that promotions are relatively infrequent business events rather than a routine pricing strategy across the dataset.

### Why It Matters

The imbalance between promotional and non-promotional periods suggests that promotional pricing introduces temporary shifts in customer purchasing behaviour rather than defining the underlying demand pattern. Consequently, promotions should be treated as an explanatory factor influencing demand variability instead of being considered the primary determinant of demand. This distinction is important because the baseline demand behaviour remains largely independent of promotional events.

### Impact on Forecasting

Since promotional periods exhibit behaviour that differs from regular selling periods, incorporating promotion-related variables into the modelling dataset was justified. Features such as **regular_price**, **price_drop_pct**, and **promo_flag** enable forecasting models to distinguish between baseline demand and demand generated by temporary pricing interventions. This improves the model's ability to capture demand variability without attributing promotional spikes to normal purchasing behaviour.

## 6.3 Discount Depth Analysis

### Purpose

While the previous analysis identified promotional periods, it does not indicate the magnitude of price reductions offered during those promotions. Examining discount depth helps distinguish between minor pricing adjustments and substantial promotional campaigns. This provides a better understanding of pricing behaviour and the potential influence of discount intensity on customer demand.

### Key Observation

The estimated promotional discounts varied across store–product combinations, indicating that promotional campaigns were implemented with different levels of pricing intensity rather than using a single standardized discount. This variation suggests that promotional activity within the dataset represents multiple pricing strategies instead of a uniform markdown policy.

### Why It Matters

Different discount levels are likely to generate different levels of customer response. Treating all promotions identically would ignore variations in promotional effectiveness and could mask meaningful demand behaviour associated with stronger or weaker price reductions. Analysing discount depth therefore provides additional context beyond a simple promotion indicator.

### Impact on Forecasting

The observed variation in discount intensity justifies representing promotional activity using both a continuous measure (**price_drop_pct**) and a categorized feature (**discount_bucket**). This enables forecasting models to distinguish between the effects of mild and aggressive promotional campaigns rather than treating every promotion as identical.

## 6.4 Store-Level Demand Analysis

### Purpose

Demand characteristics can vary considerably across retail locations due to differences in customer demographics, purchasing behaviour, and local market conditions. This section examines both the overall distribution of weekly demand across stores and the influence of promotional activity at the store level. Analysing these patterns helps determine whether demand behaviour is consistent across the retail network or varies between locations, providing important context for forecasting at the store level.

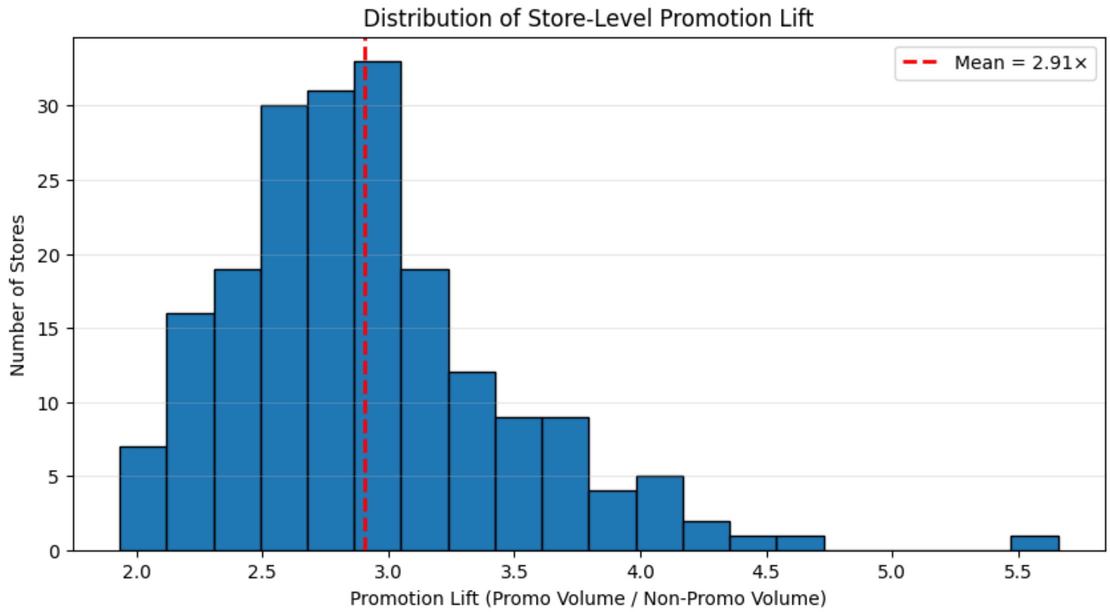


Figure X. Store-Level Comparison of Promotional and Non-Promotional Weekly Sales.

### Key Observation

The analysis shows that promotional periods generally result in higher weekly sales across most stores. However, the magnitude of this increase is not uniform. While some stores exhibit a substantial demand uplift during promotions, others experience only modest improvements. This indicates that promotional effectiveness varies across locations rather than producing a consistent response throughout the retail network.

### Why It Matters

The variation in promotional response suggests that demand is influenced by store-specific characteristics in addition to pricing strategies. Factors such as customer purchasing habits, regional demand patterns, and local competition may contribute to these differences. As a result, forecasting demand using a single aggregate behaviour would overlook important variations between stores.

### Impact on Forecasting

The observed differences in promotional responsiveness reinforce the decision to perform forecasting at the **store-product-week** level rather than at an aggregated store or product level. Maintaining store-level granularity preserves local demand behaviour and enables forecasting models to capture variations that would otherwise be lost through aggregation.

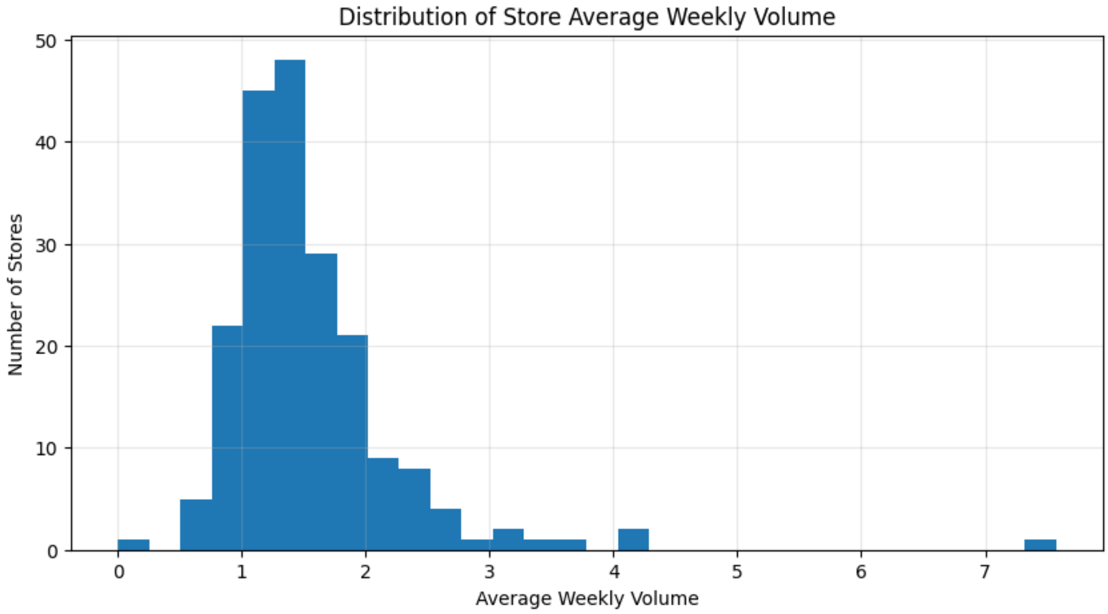


Figure X. Distribution of Average Weekly Demand Across Stores

### Observation

The distribution of average weekly demand indicates that most stores operate within a similar demand range, while a smaller number of stores exhibit noticeably higher average sales volumes. This suggests that although overall demand is relatively consistent across much of the retail network, certain stores contribute disproportionately to total sales, reflecting differences in customer traffic, purchasing behaviour, or market size.

## 6.5 Representative Demand Pattern

### Purpose

Before selecting forecasting models, it is important to understand how demand evolves over time for an individual store–product combination. Examining a representative demand series provides a visual understanding of the characteristics of intermittent demand, including prolonged zero-demand periods, irregular demand occurrences, and sudden demand spikes. This analysis helps verify whether the observed demand behaviour aligns with the assumptions of intermittent demand forecasting methods.

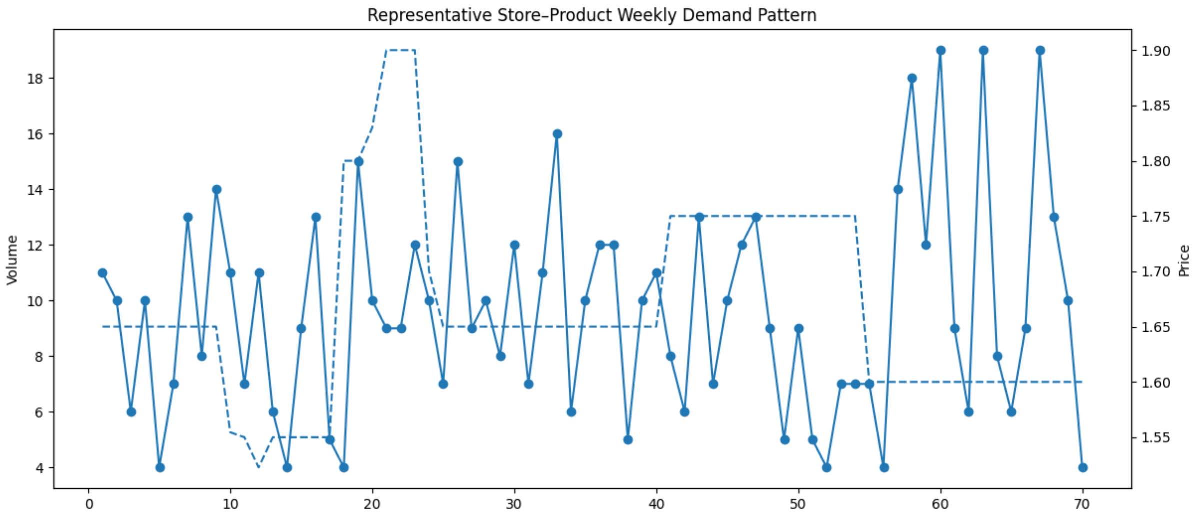


Figure X. Representative Weekly Demand Pattern for a Store–Product Combination.

### Key Observation

The representative demand series exhibits long periods of zero demand interrupted by occasional non-zero sales. Demand occurrences are irregular, and when sales do occur, the quantities vary considerably between weeks. Rather than following a smooth or seasonal trend, the series demonstrates the sparse and unpredictable behaviour that is characteristic of intermittent retail demand.

### Why It Matters

Traditional forecasting techniques generally assume that demand occurs continuously over time and that recent observations provide reliable information about future demand. The representative series demonstrates that these assumptions do not hold for many products within the dataset. The presence of irregular demand intervals and variable demand sizes makes forecasting substantially more challenging than conventional retail time-series problems.

### Impact on Forecasting

The observed demand behaviour provides strong justification for evaluating specialized intermittent demand forecasting methods such as **Croston's Method**, **Syntetos–Boylan Approximation (SBA)**, and **Teunter–Syntetos–Babai (TSB)** in addition to a traditional Moving Average benchmark. These methods are specifically designed to account for the irregular occurrence of demand and are therefore better suited for sparse retail demand series.

## 6.6 Product-Level Promotional Sensitivity

### Purpose

Promotional effectiveness is unlikely to be identical across all products. Evaluating product-level promotional sensitivity helps identify how individual products respond to price reductions and whether promotional uplift is consistent across the product portfolio. This analysis provides insight into product-specific demand behaviour before developing forecasting models.

| Masked Product ID | Non-Promo Volume | Promo Volume | Promo Lift | Total Volume |
|---|---|---|---|---|
| **8237** | 0.08237964626879789 | 2.440677966101695 | 29.627196481737336 | 1447.0 |
| **2359** | 0.1911993413228433 | 3.460240963855422 | 18.09755692626967 | 6398.0 |
| **2324** | 0.1784332688588008 | 2.4871794871794872 | 13.93898964630672 | 3281.0 |
| **9754** | 0.15380662841483259 | 1.5617977528089888 | 10.15429418683216 | 3047.0 |
| **7908** | 0.5021842732327244 | 4.837812352663838 | 9.633540137609762 | 12790.0 |
| **1124** | 0.19701108789972682 | 1.8676671214188267 | 9.48001019297664 | 3821.0 |
| **3115** | 0.22442196531791908 | 1.389261744966443 | 6.190400048401665 | 1967.0 |
| **7596** | 0.30369012147604857 | 1.8024691358024691 | 5.935224784532961 | 4559.0 |
| **3910** | 0.20136837194837506 | 1.1554054054054055 | 5.737770009391631 | 1466.0 |
| **9497** | 0.250964588870788 | 1.3157894736842106 | 5.242928811608796 | 3227.0 |
| **9244** | 0.4346917450365726 | 2.072463768115942 | 4.767663043478261 | 975.0 |
| **3494** | 0.5096842920782491 | 2.4245014245014245 | 4.756869030857251 | 7816.0 |
| **4536** | 0.4243458475540387 | 2.013333333333333 | 4.74455764075067 | 1874.0 |
| **4245** | 0.4044560091071719 | 1.6233333333333333 | 4.013621498458652 | 2974.0 |
| **2438** | 0.4292841305036427 | 1.7087719298245614 | 3.9805150211814353 | 6395.0 |
| **7358** | 0.5001836884643645 | 1.967303609341826 | 3.9331622656103193 | 10079.0 |
| **8684** | 0.4720909886264217 | 1.7420408163265306 | 3.6900531005582367 | 4832.0 |
| **6183** | 0.91098144823459 | 3.2027027027027026 | 3.5156618270430062 | 6563.0 |
| **6939** | 0.5300564061240934 | 1.7108433734939759 | 3.2276628557403835 | 3431.0 |
| **3932** | 0.545843337334934 | 1.6928104575163399 | 3.1012752959144714 | 8052.0 |

Table X. Product-Level Promotional Sensitivity Based on Promotion Lift

## Key Observation

The analysis reveals noticeable differences in promotional sensitivity across products. While certain products experience substantial increases in demand during promotional periods, others show relatively modest changes in weekly sales. This variation indicates that customer response to promotions is product-dependent rather than uniform across the entire product portfolio.

## Why It Matters

Differences in promotional sensitivity suggest that demand is influenced by product-specific characteristics such as purchasing frequency, customer preference, and price elasticity. Treating all products as responding similarly to promotions would overlook these behavioural differences and could reduce forecasting accuracy.

### Impact on Forecasting

The observed variability supports maintaining **product-level granularity** throughout the forecasting process. It also reinforces the inclusion of promotion-related variables, allowing forecasting models to capture differences in promotional response across individual products instead of assuming a common demand pattern.

## 6.7 Distribution of Product Promotion Lift

### Purpose

While the previous analysis identified individual products with the highest promotional sensitivity, it does not describe how promotional response is distributed across the entire product portfolio. Examining the distribution of promotion lift helps determine whether strong promotional effects are widespread or limited to a small subset of products. This provides a broader understanding of how promotions influence demand across different products.

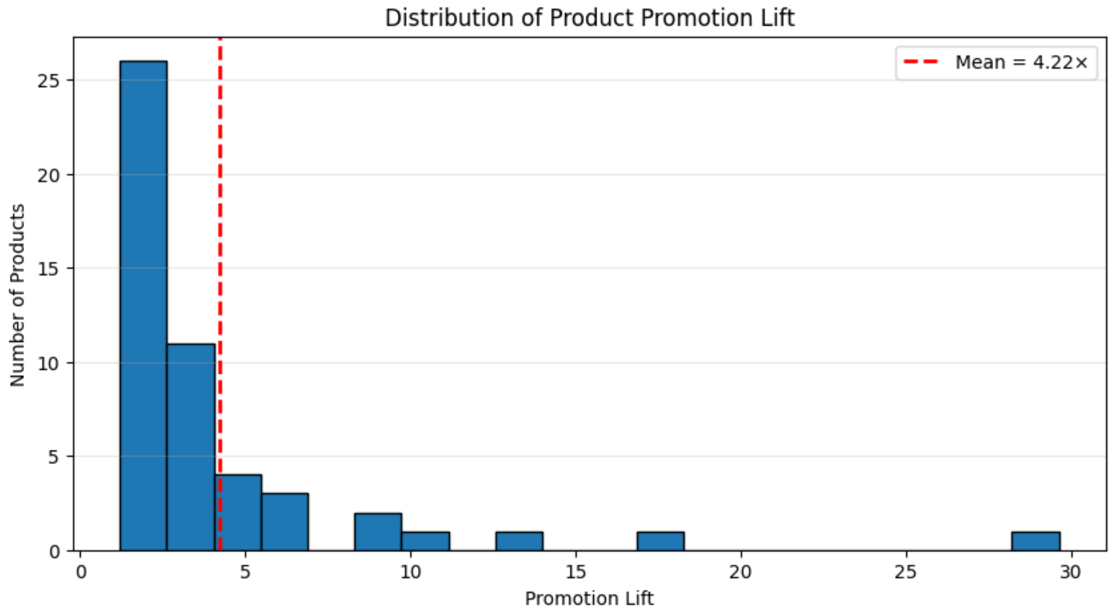


Figure X. Distribution of Product Promotion Lift Across Products

### Key Observation

The distribution of promotion lift shows that promotional response varies considerably across products. Most products exhibit relatively modest promotion lift, while a smaller number of products experience substantially larger increases in demand during promotional periods. This indicates that strong promotional responsiveness is concentrated among a subset of products rather than being uniformly observed across the product portfolio.

### Why It Matters

The uneven distribution of promotion lift demonstrates that products differ in their sensitivity to promotional pricing. Applying a uniform assumption about promotional impact would therefore fail to capture important product-specific demand behaviour. Understanding this variability is essential for interpreting promotional effectiveness and developing forecasting models that reflect real purchasing patterns.

### Impact on Forecasting

The observed distribution reinforces the need to retain product-level information throughout the forecasting pipeline. It also supports incorporating promotion-related features into the modelling dataset, enabling forecasting models to distinguish between products that exhibit strong promotional response and those whose demand remains relatively stable during promotional periods.

## 6.8 Holiday Impact Analysis

### Purpose

Retail demand is often influenced by recurring calendar events that alter consumer purchasing behaviour. To evaluate the significance of these effects within the dataset, demand and sales value were compared across four periods: **Pre-Holiday, Holiday, Post-Holiday, and Normal** weeks. This analysis determines whether holiday-related periods introduce systematic changes in demand that should be considered during forecasting.

| Holiday Period | Avg. Weekly Volume | Avg. Weekly Sales Value |
|---|---|---|
| **NORMAL** | 1.52 | 7.72 |
| **HOLIDAY** | 1.66 | 8.73 |
| **POST** | 1.57 | 8.10 |
| **PRE** | 1.61 | 8.49 |

Table X. Average Weekly Demand and Sales Value Across Holiday Periods

### Key Observation

Average weekly demand increases during holiday-related periods, reaching its highest level during holiday weeks. Demand also remains elevated during the weeks immediately before and after holidays when compared with normal trading periods. A similar pattern is observed for weekly sales value, indicating that holiday periods consistently generate stronger retail activity than non-holiday weeks.

### Why It Matters

Although the increase in demand during holiday periods is moderate rather than extreme, the pattern is consistent across the dataset. This suggests that holidays act as recurring external drivers of demand and contribute to predictable variations in customer purchasing behaviour.

Ignoring these calendar effects could lead to systematic underestimation of demand during holiday periods.

### Impact on Forecasting

The observed holiday uplift justifies the inclusion of **holiday_flag**, **holiday_window_flag**, and **holiday_period** within the modelling dataset. These features enable forecasting models to account for recurring calendar effects without attributing temporary holiday-driven demand increases to normal purchasing behaviour.

## 6.9 Demand Pattern Classification (ADI–$CV^2$)

### Purpose

Retail products exhibit different demand characteristics, making it important to understand the frequency and variability of demand before selecting forecasting methods. The ADI–$CV^2$ framework classifies each store–product time series into four demand categories—Smooth, Erratic, Intermittent, and Lumpy—based on the average interval between demand occurrences and the variability of non-zero demand. This analysis provides an objective assessment of the dominant demand patterns within the dataset and helps determine the suitability of different forecasting approaches.

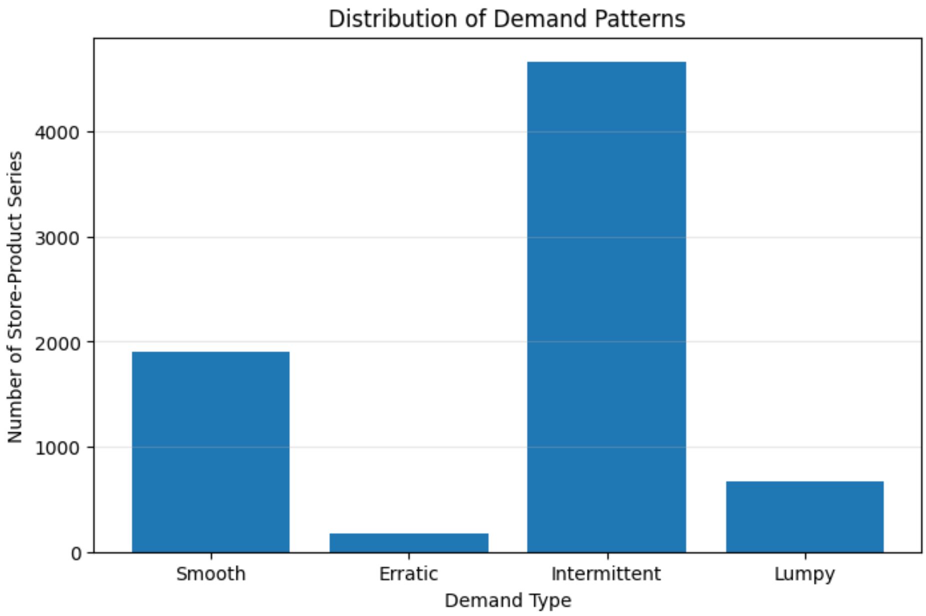


Figure X. Distribution of Store–Product Series Across ADI–$CV^2$ Demand Categories

### Key Observation

The ADI–$CV^2$ classification indicates that **Intermittent demand** is the dominant demand pattern within the dataset, representing approximately **63%** of all active store–product series. Smooth demand forms the second largest category, while Erratic and Lumpy demand patterns account for a comparatively smaller proportion of the portfolio. This distribution confirms that irregular purchasing behaviour is a defining characteristic of the retail demand data.

### Why It Matters

The predominance of intermittent demand indicates that many products experience extended periods of zero demand interspersed with occasional sales. Such behaviour differs substantially from the continuous demand patterns assumed by many conventional forecasting techniques. The coexistence of multiple demand categories also highlights the diversity of demand behaviour across products, increasing the complexity of the forecasting task.

### Impact on Forecasting

The ADI–$CV^2$ analysis provides the primary justification for evaluating specialized intermittent demand forecasting methods in this study. Since intermittent demand represents the majority of store–product series, models specifically designed for sparse demand—including **Croston's Method**, **Syntetos–Boylan Approximation (SBA)**, and **Teunter–Syntetos–Babai (TSB)**—were selected alongside a Moving Average benchmark to determine the most appropriate forecasting approach.

## 6.10 Promotion Sensitivity by Demand Pattern

### Purpose

Following the classification of demand patterns using the ADI–$CV^2$ framework, promotional response was evaluated separately for each demand category. This analysis investigates whether products with different demand characteristics exhibit similar or distinct responses to promotional activity. Understanding these differences provides additional insight into demand behaviour and helps determine whether promotional effects should be considered alongside demand intermittency during forecasting.

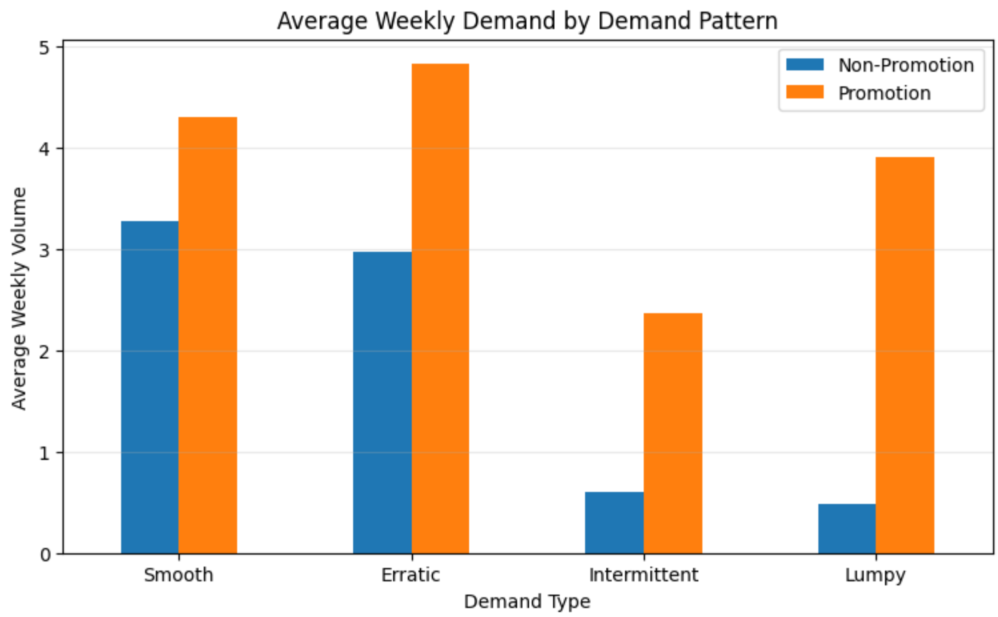


Figure X. Average Weekly Demand During Promotional and Non-Promotional Periods Across Demand Categories

## Key Observation

Promotional activity increases average demand across all demand categories; however, the magnitude of the increase differs considerably. Smooth demand series maintain the highest average sales during both promotional and non-promotional periods due to their naturally consistent purchasing behaviour. In contrast, Intermittent and Lumpy demand series exhibit a proportionally larger uplift during promotions, indicating that promotional activity can partially offset otherwise infrequent purchasing patterns.

## Why It Matters

The differing promotional response across demand categories demonstrates that demand behaviour is influenced by both **intrinsic demand characteristics** and **external promotional activity**. Products with intermittent demand are not only irregular in their purchasing patterns but also respond differently to promotions compared with products exhibiting smooth demand. This highlights the interaction between demand intermittency and promotional effects within the retail environment.

## Impact on Forecasting

The combined findings from the ADI–$CV^2$ classification and promotion sensitivity analysis support the inclusion of promotion-related variables when forecasting intermittent demand. They also reinforce the evaluation of specialized intermittent forecasting methods, ensuring that model performance is assessed under realistic demand conditions where both irregular demand occurrence and promotional influences are present.

## 6.11 Top Product Trend Analysis

### Purpose

To understand whether demand patterns were driven by individual products or reflected broader market behaviour, the weekly sales trends of the highest-selling products were analysed over time. Examining these products helps identify common demand movements, seasonal trends, and unusual demand fluctuations that may affect multiple products simultaneously.

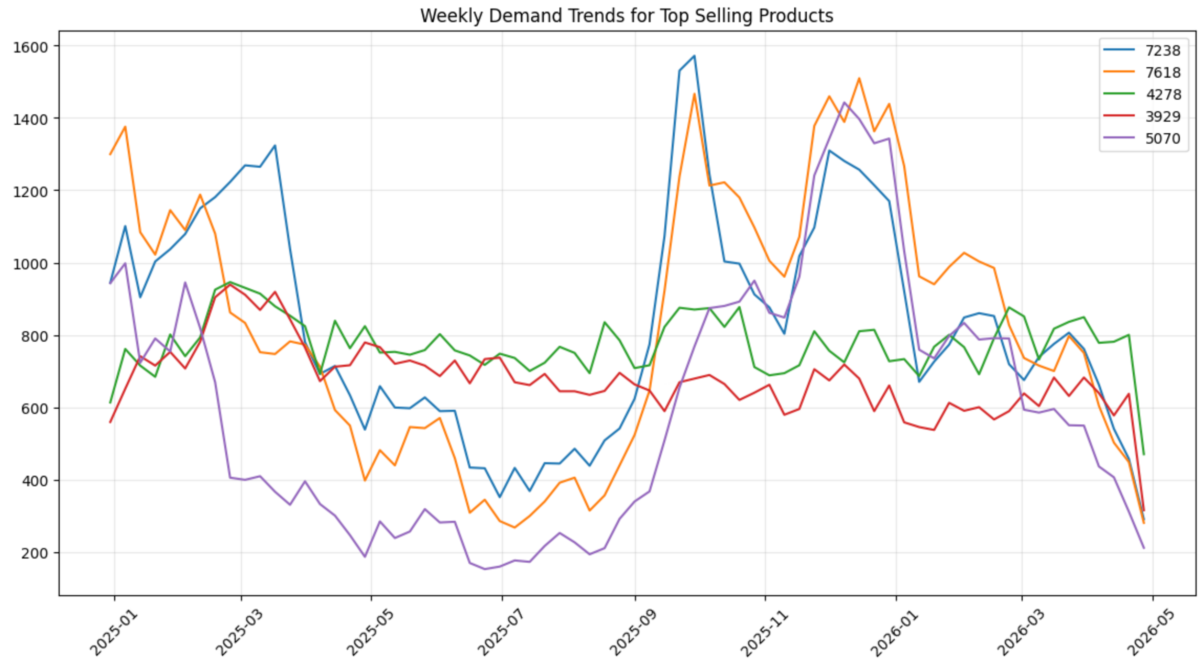


Figure X. Weekly Demand Trends of the Highest-Selling Products

### Key Observation

Although the highest-selling products differ in their overall demand levels, several products exhibit a noticeable increase in weekly demand during the same period near the end of the observation window. The simultaneous nature of this increase across multiple products suggests that the demand surge is unlikely to be product-specific and instead reflects a broader external influence affecting retail demand.

### Why It Matters

When multiple products display similar demand movements over the same time period, the underlying driver is more likely to be a common external factor rather than changes in the behaviour of individual products. Identifying these synchronized demand patterns helps distinguish between product-level variability and broader market events that influence purchasing behaviour across the retail portfolio.

## Impact on Forecasting

The synchronized increase in demand motivated a focused investigation of the year-end period to determine whether the observed pattern was associated with seasonal events, holidays, promotional activity, or a combination of these factors. Understanding the source of this demand surge is important for ensuring that forecasting models correctly interpret recurring calendar-driven demand rather than treating it as random variation.

# 6.12 Year-End Spike Investigation

## Purpose

The previous analysis identified a synchronized increase in demand across several high-selling products during the latter part of the observation period. To determine whether this increase was associated with promotional activity, weekly demand was analysed alongside the number of products under promotion. Comparing these two trends helps assess the extent to which promotional campaigns contributed to changes in aggregate retail demand.

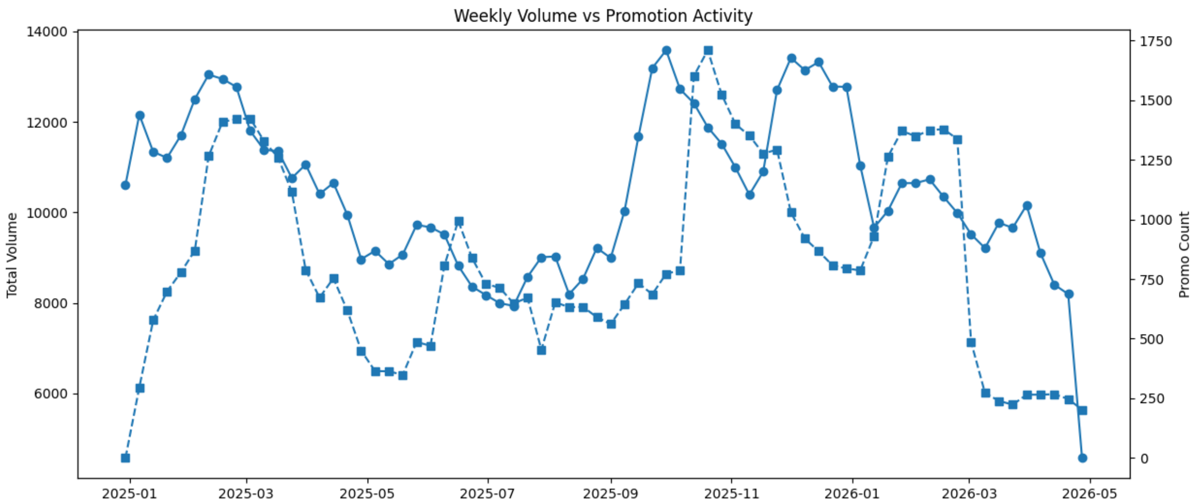


Figure X. Weekly Demand and Promotional Activity Throughout the Observation Period

## Key Observation

Weekly demand and promotional activity exhibit similar movement over time, with several periods of increased promotional activity coinciding with higher aggregate demand. However, the relationship is not perfectly consistent, indicating that while promotions contribute to demand growth, they are not the sole driver of fluctuations in weekly sales. Other external factors, including seasonal demand patterns, are also likely to influence purchasing behaviour.

## Why It Matters

The observed relationship demonstrates that promotional campaigns play an important role in shaping retail demand but should not be considered in isolation. Demand fluctuations result from the combined influence of promotions, seasonal effects, and underlying consumer purchasing patterns. Understanding these interacting factors provides a more realistic representation of retail demand behaviour.

### Impact on Forecasting

The analysis supports retaining promotion-related features within the forecasting dataset while also considering calendar-based variables that capture seasonal demand patterns. Incorporating both sources of information enables forecasting models to distinguish temporary promotional uplift from broader market-driven demand changes.

## 6.13 Christmas & New Year Surge

### Purpose

Following the investigation of promotional activity, the year-end period was examined in greater detail to determine whether the observed demand increase coincided with major holiday events. Weekly aggregate demand was analysed around the Christmas and New Year period to identify recurring seasonal effects that may influence retail purchasing behaviour.

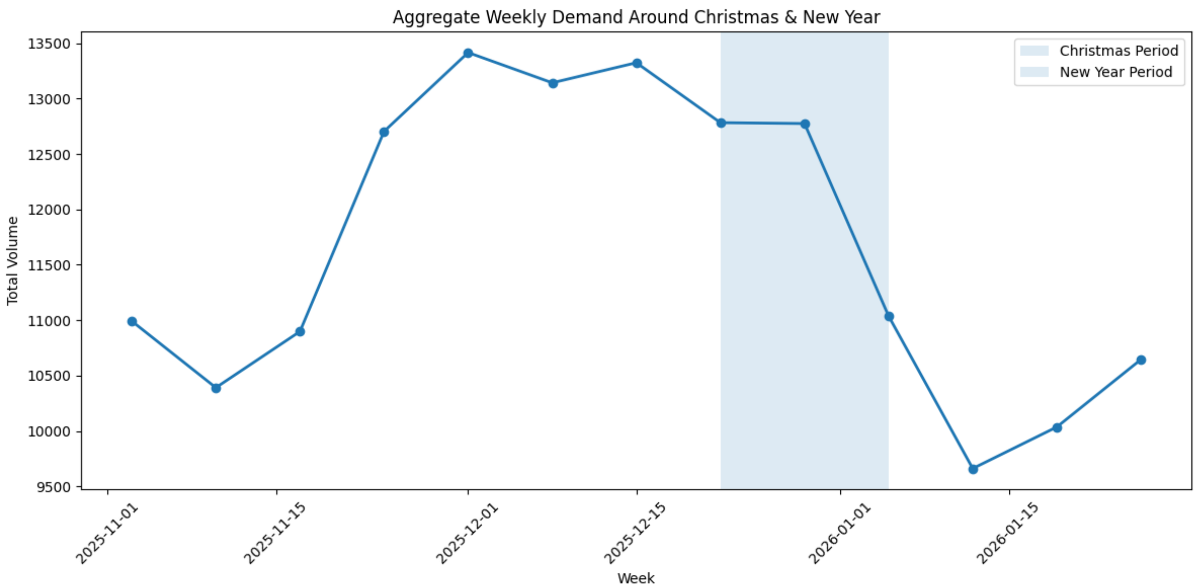


Figure X. Aggregate Weekly Demand During the Christmas and New Year Period

### Key Observation

Aggregate weekly demand increases noticeably during the weeks leading up to and including the Christmas and New Year period before declining in the subsequent weeks. The timing of this demand surge aligns closely with major holiday events, indicating that consumer purchasing activity intensifies during the year-end season. Unlike the broader promotion analysis, this pattern is concentrated within a specific calendar period and is consistently observed across the aggregated demand series.

### Why It Matters

The year-end surge represents a predictable seasonal demand pattern rather than a random fluctuation. Retail businesses typically experience increased purchasing activity during holiday periods due to gifting, festive celebrations, and advance stocking by consumers. Recognizing these recurring calendar-driven patterns is essential for accurate inventory planning and demand forecasting.

### Impact on Forecasting

The identified Christmas and New Year demand surge reinforces the inclusion of calendar-based features such as **holiday_flag**, **holiday_window_flag**, and **holiday_period** within the forecasting dataset. These variables enable forecasting models to account for recurring seasonal demand increases and improve prediction accuracy during important holiday periods.

## 6.14 Forecastability Analysis

### Purpose

Forecastability refers to the ease with which future demand can be predicted based on historical observations. To evaluate the forecasting complexity of the retail demand series, the coefficient of variation (CV) was analysed across store–product combinations. The CV measures demand variability relative to its average level, providing an indication of demand stability and expected forecasting difficulty. Series with lower CV values are generally more predictable, whereas higher CV values indicate greater uncertainty and increased forecasting complexity.

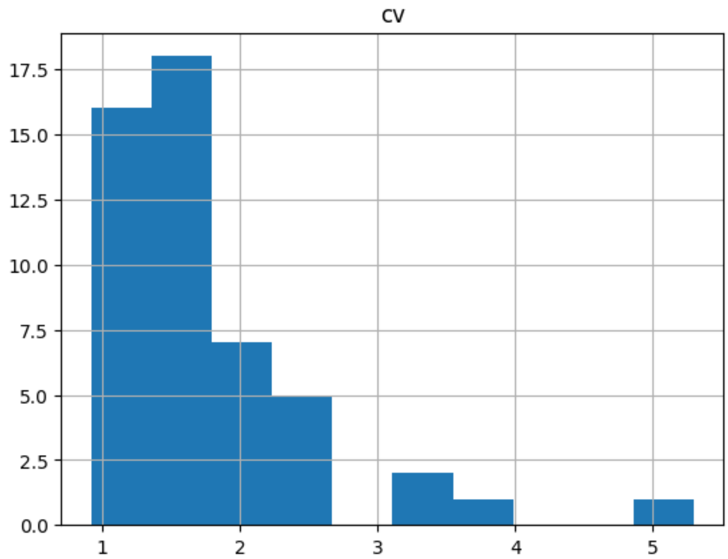


Figure X. Distribution of the Coefficient of Variation (CV) Across Store–Product Series

### Key Observation

The coefficient of variation is concentrated at relatively lower values for most store–product series, indicating that a large proportion of products exhibit moderate demand variability. However, a smaller number of series display substantially higher CV values, representing products with highly variable demand that are inherently more difficult to forecast. This demonstrates that forecasting complexity varies across the retail portfolio rather than being uniform for all products.

### Why It Matters

Differences in demand variability directly influence forecasting performance. Stable demand series generally allow more reliable predictions, whereas highly variable series are associated with greater forecasting uncertainty. Recognizing this variation helps explain why forecasting accuracy may differ across products even when the same forecasting method is applied.

### Impact on Forecasting

The forecastabilityt analysis reinforces the need to evaluate forecasting models across a diverse set of demand series rather than assuming consistent performance for the entire product portfolio. Combined with the ADI–$CV^2$ classification, the CV analysis confirms that both demand intermittency and demand variability must be considered when selecting and evaluating forecasting methods.

## 6.15 Store Promotion Rate Analysis

### Purpose

Promotional activity is not necessarily applied uniformly across retail locations. To understand the consistency of promotional strategies throughout the retail network, the promotion rate for each store was analysed. Examining the distribution of store-level promotion rates helps identify whether stores follow a similar promotional strategy or whether substantial differences exist across locations.

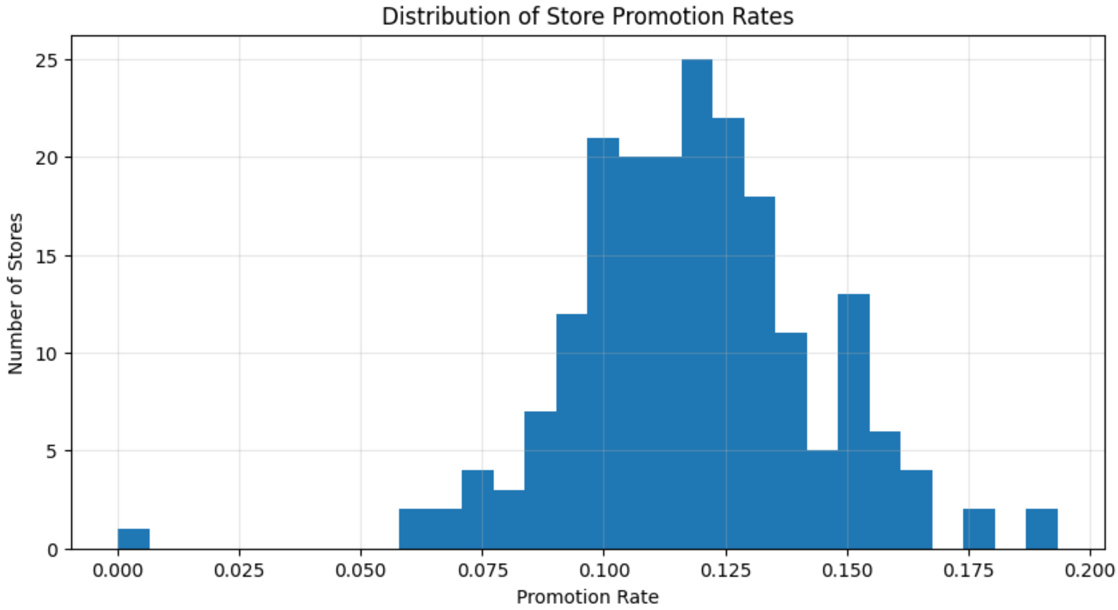


Figure X. Distribution of Promotion Rates Across Stores

## Key Observation

Most stores exhibit promotion rates within a relatively narrow range, indicating that promotional campaigns are implemented with reasonable consistency across the retail network. However, a small number of stores display noticeably lower or higher promotion rates, suggesting localized promotional strategies or differences in store operations. Overall, the distribution indicates that while promotional practices are broadly standardized, some variability exists between stores.

## Why It Matters

Consistent promotion rates reduce the likelihood that forecasting differences are driven solely by unequal promotional exposure across stores. At the same time, the presence of a small number of stores with atypical promotion rates highlights the importance of preserving store-level information during modelling, as localized promotional behaviour may still influence demand patterns.

## Impact on Forecasting

The relatively consistent distribution of promotion rates supports the use of a common forecasting framework across stores while retaining store-level granularity. Including promotion-related features enables forecasting models to capture the influence of localized promotional activity without assuming identical promotional behaviour across the entire retail network.

## 6.16 Product Promotion Frequency Analysis

### Purpose

While the previous analysis examined promotional activity across stores, this section evaluates how frequently individual products are promoted throughout the observation period. Understanding product-level promotion frequency helps determine whether promotional campaigns are distributed evenly across the product portfolio or concentrated on a subset of products. This provides additional context for interpreting demand patterns and promotional effects observed during the exploratory analysis.

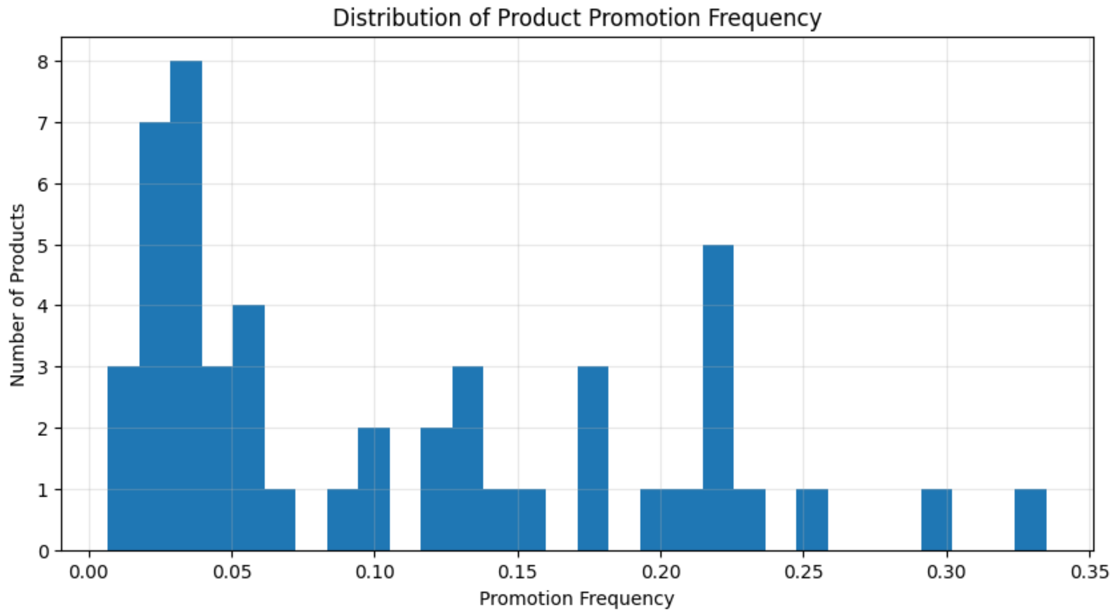


Figure X. Distribution of Promotion Frequency Across Products

### Key Observation

The distribution indicates considerable variation in product promotion frequency. While many products are promoted relatively infrequently, a smaller group of products experiences promotional campaigns much more often. This suggests that promotional efforts are concentrated on selected products rather than being applied uniformly across the entire product portfolio.

### Why It Matters

Differences in promotion frequency can lead to substantial variation in historical demand patterns between products. Frequently promoted products are more likely to exhibit repeated demand spikes, whereas products with limited promotional exposure tend to display demand driven primarily by baseline purchasing behaviour. Recognizing this distinction is important when interpreting promotional lift and evaluating forecasting performance across different products.

### Impact on Forecasting

The observed variation in promotion frequency reinforces the inclusion of promotion-related variables within the forecasting dataset. Features such as **promo_flag**, **price_drop_pct**, and **discount_bucket** enable forecasting models to distinguish between baseline demand and promotion-driven demand, improving their ability to predict future sales under varying promotional conditions.

## 6.17 Summary of Key EDA Findings

### Section Objective

The exploratory analysis provided a comprehensive understanding of the retail demand data and identified the key characteristics that influenced the forecasting methodology adopted in this study. Rather than serving as descriptive statistics alone, the EDA established evidence-based insights that guided data preparation, feature engineering, forecasting granularity, and model selection.

### Key Findings

- The dataset is dominated by **intermittent demand**, with approximately 63% of store–product series classified as intermittent using the ADI–$CV^2$ framework.
- Promotional activity consistently increases demand, although the magnitude of promotional uplift varies across products, stores, and demand patterns.
- Holiday periods, particularly Christmas and New Year, produce recurring increases in weekly demand that represent predictable seasonal behaviour rather than random fluctuations.
- Demand characteristics differ across stores and products, supporting the decision to preserve forecasting at the **store–product–week** level instead of using aggregated data.
- Product promotion frequency and store promotion rates exhibit measurable variation, demonstrating that promotional exposure is not identical across the retail network.
- Forecastability differs across demand series due to variations in demand variability, indicating that forecasting difficulty is not uniform across products.
- Calendar effects, promotional activity, and demand intermittency collectively influence retail demand and should be incorporated into the forecasting framework.

### Implications for Modelling

The findings of the exploratory analysis directly informed the forecasting methodology implemented in this project. The dominance of intermittent demand justified the evaluation of specialized forecasting methods such as Croston, SBA, and TSB alongside a Moving Average benchmark. Observed promotional and holiday effects motivated the inclusion of promotion- and calendar-related features, while the variation in demand behaviour across stores and products supported forecasting at the store–product–week level. Together, these findings established the analytical foundation for the modelling and evaluation framework presented in the following chapter.

# 7. Forecasting Methodology

### Section Objective

This chapter describes the methodology adopted to develop and evaluate forecasting models for intermittent retail demand. Building upon the insights obtained from the exploratory data analysis, a dedicated modelling dataset was prepared, a standardized forecasting framework was established, and multiple forecasting techniques were implemented and compared. The objective of this chapter is to describe the forecasting pipeline, justify the selected modelling approaches, and present the evaluation methodology used to identify the most suitable forecasting model for intermittent demand.

## 7.1 Modelling Dataset Preparation

### Purpose

The weekly aggregated dataset developed during the data preparation stage provided the foundation for exploratory analysis but did not contain all the information required for forecasting. To support model development, a dedicated modelling dataset was created by extending the weekly dataset with additional variables describing pricing behaviour, promotional activity, and holiday effects. Integrating these variables into a single time-series dataset ensured that every forecasting method was evaluated using a consistent and comprehensive representation of historical demand.

### Dataset Development

The modelling dataset, named **`weekly_modelling_dataset.csv`**, was derived from the original transactional dataset **`masked_sales_export.csv`** through weekly aggregation and subsequent feature generation. Each record represents the weekly demand of a specific product at a particular store, uniquely identified by the combination of **`week_start`**, **`masked_store_id`**, and **`masked_product_id`**. The resulting dataset serves as the common input for all forecasting models evaluated in this study.

| Variable | Description |
|---|---|
| **week_start** | Start date of the corresponding week |
| **week_id** | Sequential week identifier |
| **masked_store_id** | Anonymized store identifier |
| **masked_product_id** | Anonymized product identifier |
| **weekly_volume** | Total weekly sales volume (forecast target) |
| **weekly_value** | Total weekly sales value |
| **weekly_avg_price** | Average selling price during the week |

| | |
|---|---|
| **regular_price** | Estimated regular selling price |
| **price_drop_pct** | Percentage price reduction from the regular price |
| **promo_flag** | Indicator of promotional activity |
| **discount_bucket** | Categorized discount depth |
| **holiday_flag** | Indicator of holiday weeks |
| **holiday_window_flag** | Indicator of weeks surrounding holidays |
| **holiday_period** | Classification of Normal, Pre-Holiday, Holiday, or Post-Holiday period |

Table X. Variables Included in the Weekly Modelling Dataset

### Why It Matters

Preparing a dedicated modelling dataset ensures that all forecasting methods operate on the same standardized input. By combining historical demand with pricing, promotional, and calendar-related information, the dataset provides a comprehensive representation of the factors identified during the exploratory analysis as influencing retail demand. This consistency enables fair comparison between forecasting methods while maintaining a reproducible modelling pipeline.

### Transition to Forecasting Framework

With the modelling dataset prepared, the next step was to establish a consistent forecasting framework. This included defining the forecasting objective, selecting the forecasting granularity, specifying the training and testing periods, and implementing a rolling-origin evaluation strategy to ensure that model performance reflected real-world forecasting conditions.

## 7.2 Forecasting Framework

### Purpose

A consistent forecasting framework was established to ensure that all forecasting methods were evaluated under identical conditions. The framework defines the forecasting objective, the level of forecasting granularity, the train–test partitioning strategy, and the evaluation methodology. Using a common framework ensures that differences in forecasting performance are attributable to the forecasting methods themselves rather than inconsistencies in the evaluation process.

### Forecasting Objective

The primary objective of the forecasting framework is to predict the **weekly sales volume (`weekly_volume`)** for each unique store–product combination. For every forecasting step, the model estimates demand for the upcoming week using only information that would have been available at the time the prediction was made. This objective closely reflects real-world retail forecasting, where future demand must be estimated without access to future observations.

## Forecasting Granularity

Forecasting was performed at the **store–product–week** level, where each observation is uniquely identified by the combination of **`week_start`**, **`masked_store_id`**, and **`masked_product_id`**. This level of granularity preserves the demand behaviour of individual products within each store while maintaining the weekly temporal resolution established during data preparation. Forecasting at this level enables the models to capture localized demand patterns that would otherwise be obscured through aggregation.

## Training and Testing Strategy

The available historical data were divided chronologically into separate training and testing periods. Observations from **Weeks 1–56** were used to establish the historical demand patterns required for forecasting, while **Weeks 57–70** were reserved for model evaluation. This chronological split preserves the natural temporal ordering of the data and prevents future observations from influencing historical forecasts.

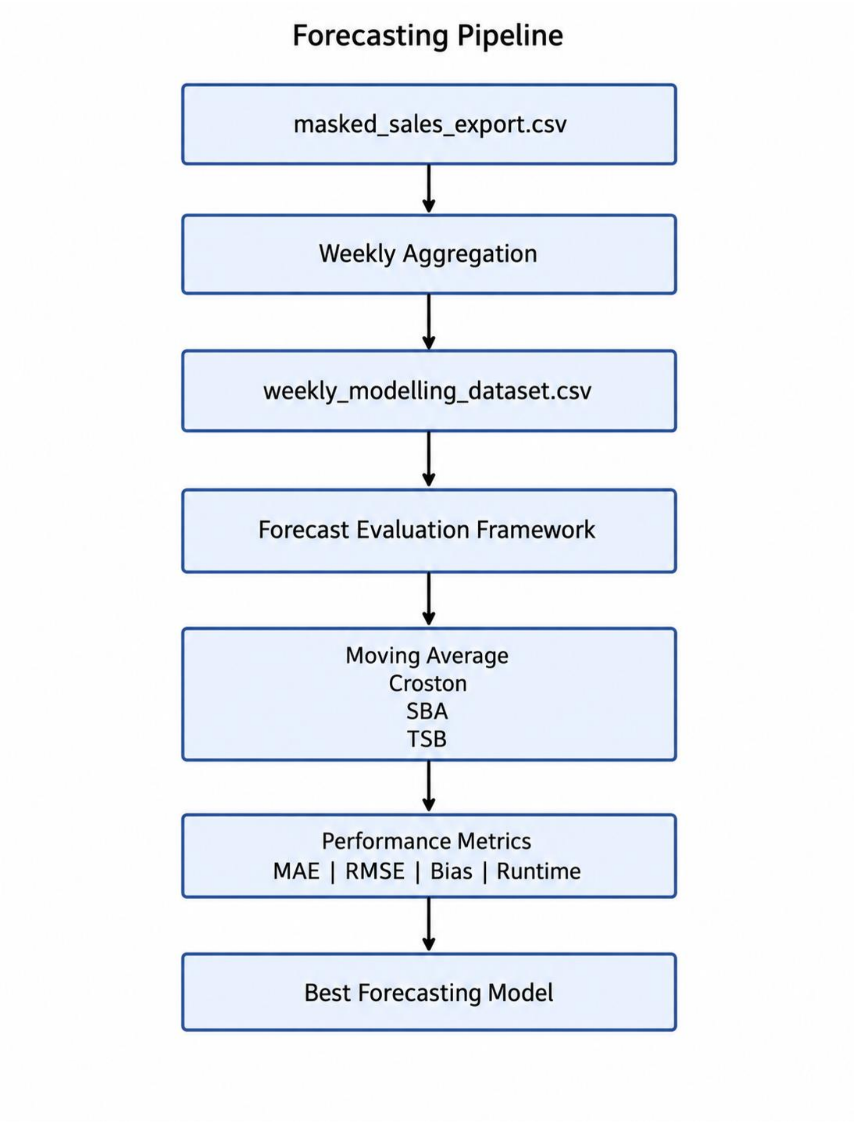


Figure X. Overall Forecasting Pipeline Adopted in This Study

The forecasting pipeline summarizes the complete workflow adopted in this study. Historical transactional data were first aggregated into weekly observations and transformed into the modelling dataset through feature generation. The prepared dataset was then evaluated using a standardized forecasting framework in which multiple forecasting methods were implemented and compared using identical performance metrics. This pipeline ensured consistency throughout the forecasting process and enabled an objective comparison between competing forecasting techniques.

## Rolling-Origin Evaluation

Model performance was evaluated using a **rolling-origin (walk-forward)** forecasting strategy. Rather than generating forecasts for the entire testing period using a fixed historical window, forecasts were produced sequentially one week at a time. For each prediction, only observations available before the forecast week were used. After the actual demand for a week became available, it was incorporated into the historical data before forecasting the subsequent week. This procedure mirrors how forecasting systems operate in practice and provides an unbiased assessment of model performance under realistic operating conditions.

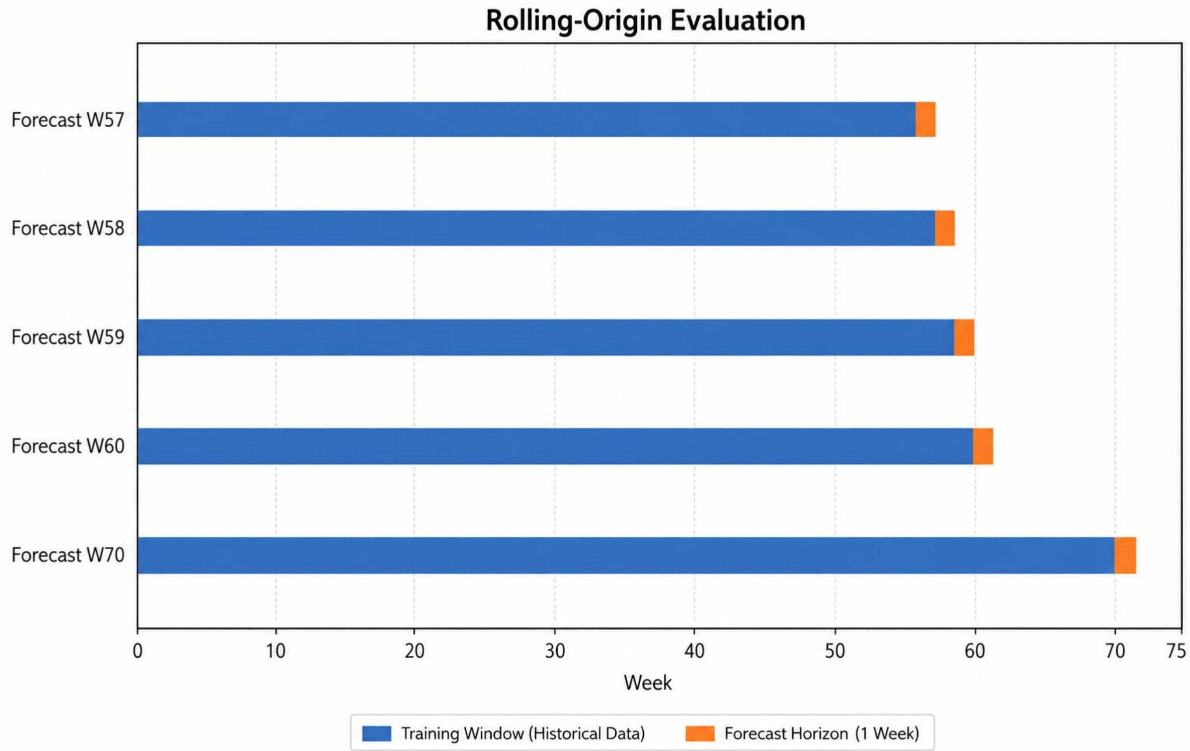


Figure X. Rolling-Origin (Walk-Forward) Forecast Evaluation Strategy

The rolling-origin evaluation strategy ensures that each forecast is generated using only historical observations available prior to the prediction period. After each forecast is evaluated against the actual demand, the newly observed week is incorporated into the historical dataset before generating the next forecast. This process closely reflects real-world forecasting operations and prevents information leakage during model evaluation.

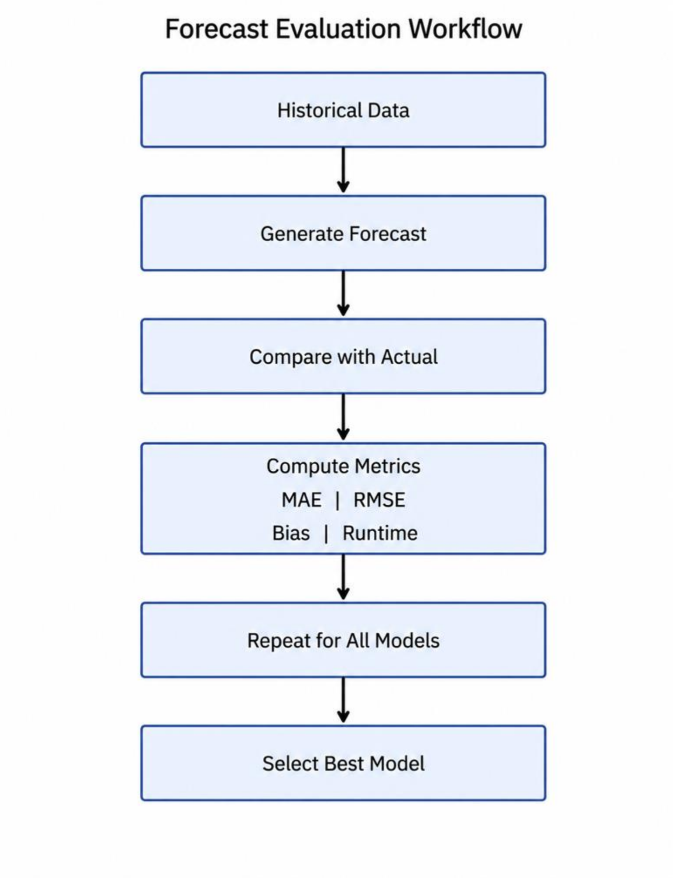


Figure X. Common Evaluation Workflow Applied to All Forecasting Models

Every forecasting model evaluated in this study followed the same evaluation workflow. Forecasts were generated using historical demand, compared against the observed values, and assessed using a common set of performance metrics including MAE, RMSE, Mean Residual, Relative Bias, and execution time. Applying an identical evaluation procedure ensured that differences in performance reflected the forecasting methods themselves rather than variations in the evaluation process.

### Why It Matters

Forecast evaluation must accurately represent the conditions under which forecasts are generated in practice. The rolling-origin strategy ensures that each prediction is based solely on historical information available at the prediction time, preventing information leakage and providing a realistic assessment of forecasting accuracy. Using the same evaluation framework for every forecasting method also guarantees a fair comparison between competing approaches.

## 7.3 Moving Average Forecasting

### Purpose

The Moving Average (MA) forecasting model was implemented as the baseline forecasting approach for this study. Due to its simplicity, computational efficiency, and widespread use in demand forecasting applications, it provides an appropriate benchmark against which specialised intermittent-demand forecasting methods can be evaluated. Establishing a strong baseline enables the relative performance of more advanced forecasting techniques to be assessed under identical evaluation conditions.

## Methodology

The Moving Average method estimates future demand by calculating the arithmetic mean of a fixed number of previous observations. Unlike intermittent-demand forecasting methods that explicitly model demand occurrence and demand size separately, the Moving Average model assumes that future demand can be represented by recent historical averages.

To identify the most appropriate historical window for the retail dataset, five window sizes were evaluated: **4, 8, 12, 26, and 52 weeks**. Shorter windows provide greater responsiveness to recent demand changes, whereas longer windows produce smoother forecasts by incorporating more historical information. Evaluating multiple window lengths allowed the influence of historical memory on forecasting accuracy to be systematically assessed.

Each Moving Average configuration was evaluated using the rolling-origin forecasting framework described in Section 7.2. For every forecast week, predictions were generated using only observations available prior to that week, ensuring that future information was never incorporated into the forecasting process.

## Implementation

The Moving Average forecasting algorithm was implemented independently for every store-product demand series. Historical observations were first grouped according to the forecasting granularity defined by the combination of **week_start**, **masked_store_id**, and **masked_product_id**. Within each series, observations were ordered chronologically before calculating rolling historical averages.

For every evaluation week, the forecast was computed as the average demand observed during the specified historical window. This process was repeated sequentially across the evaluation horizon for each Moving Average window size. Forecast accuracy was subsequently measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Residual, Relative Bias, and execution time to enable direct comparison with the intermittent-demand forecasting methods evaluated later in this study.

## Transition

The performance of each Moving Average window was subsequently compared to determine the optimal historical window size. The experimental results are presented in the following section before comparing the Moving Average approach with specialised intermittent-demand forecasting methods.

## 7.4 Moving Average Results

### Purpose

The Moving Average forecasting method was initially evaluated using multiple historical window sizes to determine the amount of historical information that produced the most accurate forecasts. Comparing several window lengths enabled the influence of historical memory on forecasting performance to be assessed and provided an objective basis for selecting the Moving Average configuration used in the final benchmarking study.

### Performance Comparison

| Moving Average Window | MAE | RMSE | Mean Residual | Relative Bias |
|---|---|---|---|---|
| **4** | **0.9428** | **1.5460** | **-0.0904** | **0.0661** |
| **8** | 0.9567 | 1.5682 | -0.1525 | 0.1114 |
| **13** | 0.9903 | 1.6211 | -0.2148 | 0.1570 |
| **26** | 1.0534 | 1.6708 | -0.2837 | 0.2073 |

Table X. Performance Comparison of Moving Average Window Sizes

### Key Findings

The experimental comparison demonstrates that forecasting accuracy is influenced by the amount of historical information included in the Moving Average calculation. Shorter historical windows responded more effectively to recent changes in demand, whereas larger windows produced smoother forecasts but were less responsive to changing demand patterns. This behaviour is consistent with the intermittent demand characteristics identified during the exploratory data analysis.

Among the evaluated configurations, the **4-week Moving Average (MA(4))** achieved the strongest overall forecasting performance and was therefore selected as the representative Moving Average model for subsequent comparison with Croston's Method, SBA, and TSB.

## Rolling-Origin Forecast Performance

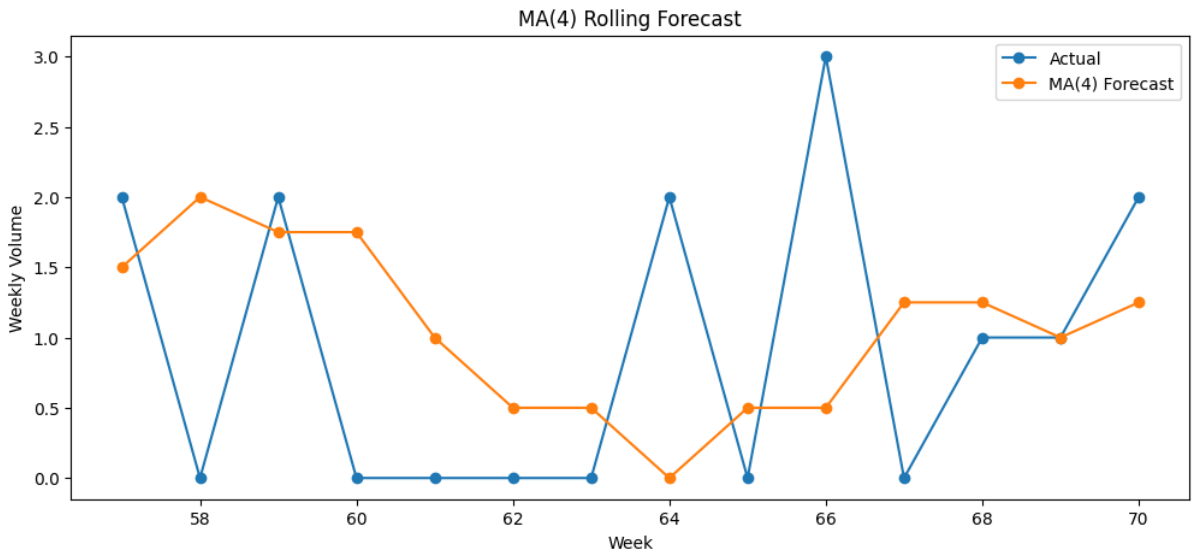


Figure X. Actual vs Forecast Weekly Demand Using the MA(4) Model

## Graph Interpretation

The rolling-origin forecast demonstrates that the MA(4) model successfully captures the overall direction of weekly demand while smoothing short-term fluctuations. Forecasts closely follow the underlying demand trend throughout the evaluation period and adapt to gradual changes in purchasing behaviour. Although sudden promotional spikes are only partially captured due to the averaging mechanism, the model provides stable and consistent short-term forecasts without introducing excessive variability.

## Model Selection

Based on the comparative evaluation, the **4-week Moving Average (MA(4))** was selected as the final Moving Average configuration. This model achieved the most favourable balance between forecasting accuracy, responsiveness to recent demand changes, and computational efficiency. Consequently, MA(4) was carried forward as the baseline forecasting model for comparison against specialised intermittent-demand forecasting methods.

## Transition

While the Moving Average model provides an effective baseline, it does not explicitly account for the intermittent occurrence of demand. The following section therefore evaluates **Croston's Method**, which was specifically developed to forecast intermittent demand by separately estimating demand size and demand intervals.

## 7.5 Croston's Method

### Purpose

Croston's Method was implemented to address the intermittent demand characteristics identified during the exploratory data analysis. Unlike the Moving Average approach, which directly averages historical observations, Croston's Method separates the forecasting process into two components: demand size and the interval between successive non-zero demand occurrences. This decomposition enables the method to better represent demand patterns containing frequent zero-demand periods, making it one of the most widely recognised forecasting techniques for intermittent demand.

### Methodology

Croston's Method estimates future demand by independently smoothing two time series: the magnitude of non-zero demand and the time interval between consecutive demand occurrences. Exponential smoothing is applied separately to both components, after which the forecast is obtained by dividing the estimated demand size by the estimated demand interval.

Similar to the Moving Average model, forecasts were generated using the rolling-origin evaluation framework described in Section 7.2. At every forecasting step, only historical observations available before the prediction week were used, ensuring that future demand information was never incorporated into the forecasting process. This evaluation strategy enabled a fair comparison between Croston's Method and the remaining forecasting approaches.

### Implementation

The Croston forecasting algorithm was implemented independently for each store-product demand series after grouping observations according to **week_start**, **masked_store_id**, and **masked_product_id**. Each series was processed chronologically to identify non-zero demand occurrences and calculate the intervals between successive demand events.

Separate exponential smoothing updates were maintained for demand magnitude and demand interval throughout the rolling-origin evaluation period. One-step-ahead forecasts were generated sequentially for every evaluation week and subsequently assessed using the same performance metrics employed throughout the study, namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Residual, Relative Bias, and execution time. This ensured methodological consistency across all forecasting models.

### Transition

The forecasting performance of Croston's Method is evaluated in the following section and compared with the Moving Average baseline to determine whether explicitly modelling intermittent demand improves forecasting accuracy.

## 7.6 Croston Results

### Purpose

Croston's Method was evaluated to determine whether explicitly modelling intermittent demand occurrences improves forecasting accuracy relative to the Moving Average baseline. The model was assessed using the same rolling-origin evaluation framework and performance metrics to ensure a fair comparison across forecasting approaches.

### Results

| Metric | Value |
|---|---|
| **MAE** | **1.168** |
| **RMSE** | **1.738** |
| **Mean Residual** | **-0.365** |
| **Relative Bias** | **0.267** |
| **Runtime (s)** | **7.65** |

Table X. Performance Metrics for Croston's Method

### Key Findings

Croston's Method produced higher forecasting errors than the Moving Average baseline across all evaluation metrics. The model recorded an RMSE of **1.738** and an MAE of **1.168**, both higher than the corresponding MA(4) values. The larger Relative Bias and Mean Residual also indicate a greater tendency towards systematic forecasting error throughout the evaluation period.

### Discussion

Although Croston's Method was specifically designed for intermittent demand forecasting, its performance on this retail dataset was inferior to the Moving Average baseline. The results suggest that the demand patterns observed in this study were better represented by recent historical demand than by separately modelling demand size and demand intervals. Furthermore, Croston's Method required substantially greater computational time, reducing its practical efficiency despite its specialised design.

### Transition

Croston's Method is known to exhibit forecasting bias under certain demand conditions. To address this limitation, the following section evaluates the **Syntetos–Boylan Approximation (SBA)**, which introduces a bias correction while retaining the fundamental principles of Croston's forecasting approach.

## 7.7 SBA Results

### Purpose

The Syntetos–Boylan Approximation (SBA) was evaluated as an enhancement to Croston's Method that reduces the positive bias commonly associated with intermittent demand forecasting. SBA was assessed using the same rolling-origin evaluation framework and performance metrics to determine whether the bias correction improves forecasting accuracy for the retail demand dataset.

### Results

| Metric | Value |
|---|---|
| **MAE** | **1.138** |
| **RMSE** | **1.699** |
| **Mean Residual** | **-0.279** |
| **Relative Bias** | **0.204** |
| **Runtime (s)** | **6.83** |

Table X. Performance Metrics for SBA

### Key Findings

Compared with Croston's Method, SBA achieved lower forecasting errors across all evaluation metrics. The RMSE decreased from **1.738** to **1.699**, while MAE and Relative Bias also improved, indicating that the bias correction contributed to more accurate and less systematically biased forecasts. Despite these improvements, SBA remained less accurate than both the Moving Average baseline and the TSB method.

### Discussion

The SBA correction successfully reduced the bias observed in Croston's forecasts and produced a measurable improvement in overall forecasting performance. However, the gains were not sufficient to outperform the simpler Moving Average approach or the TSB model. These results suggest that although correcting Croston's inherent bias improves forecast quality, the demand characteristics within this dataset are better captured using either short-term historical averaging or demand-occurrence modelling techniques. SBA also required substantially longer execution time than the Moving Average baseline, reducing its computational efficiency.

### Transition

While SBA applies a static bias correction to Croston's forecasts, the **Teunter–Syntetos–Babai (TSB)** method explicitly models both demand occurrence and demand size over time. The following section evaluates whether this more adaptive forecasting approach provides additional improvements in forecasting accuracy.

## 7.8 Teunter–Syntetos–Babai (TSB) Results

### Purpose

The Teunter–Syntetos–Babai (TSB) method was evaluated as an extension of Croston-based forecasting techniques that explicitly models both demand occurrence and demand magnitude using exponential smoothing. Unlike Croston's Method and SBA, which estimate demand intervals, TSB continuously updates the probability of demand occurrence, allowing the model to respond more effectively to changing demand patterns. The objective of this evaluation was to determine whether this adaptive approach improves forecasting accuracy for intermittent retail demand.

### Results

| Metric | Value |
|---|---|
| **MAE** | **0.981** |
| **RMSE** | **1.548** |
| **Mean Residual** | **-0.208** |
| **Relative Bias** | **0.152** |
| **Runtime (s)** | **7.08** |

Table X. Performance Metrics for the TSB Method

### Key Findings

Among the specialised intermittent demand forecasting methods evaluated, the TSB model achieved the strongest overall forecasting performance. Its RMSE of **1.548** was substantially lower than both Croston's Method and SBA, while the Mean Absolute Error and Relative Bias were also reduced. These results indicate that explicitly modelling demand occurrence enables TSB to adapt more effectively to intermittent demand patterns than the earlier Croston-based approaches.

Although TSB produced forecasting accuracy comparable to the Moving Average baseline, it required considerably greater computational time due to the additional smoothing updates performed throughout the rolling-origin evaluation process.

### Discussion

The TSB model demonstrated that incorporating demand occurrence probabilities improves forecasting performance for intermittent demand compared with traditional Croston-based methods. The adaptive updating mechanism enables the model to respond more effectively to changing demand behaviour while reducing the systematic forecasting bias observed in Croston's Method.

Despite these improvements, the Moving Average baseline continued to achieve a marginally lower RMSE while requiring only a fraction of the computational time. Consequently, although TSB proved to be the strongest specialised intermittent-demand forecasting method, its additional computational complexity did not translate into sufficient forecasting improvement to justify replacing the simpler Moving Average model for this dataset.

### Transition

The forecasting performance of all evaluated methods is compared in the following section to identify the most suitable forecasting approach based on forecasting accuracy, bias, computational efficiency, and overall practicality.

## 7.9 Comparative Model Evaluation

### Purpose

The final stage of the modelling process involved comparing the forecasting performance of all evaluated methods using a common set of accuracy, bias, and computational performance metrics. Since every forecasting model was implemented using the same modelling dataset and evaluated under the identical rolling-origin framework, the comparison provides an objective assessment of each model's suitability for forecasting intermittent retail demand.

### Overall Model Performance

| Model | MAE | RMSE | Mean Residual | Relative Bias | Runtime (s) |
|---|---|---|---|---|---|
| **MA(4)** | **0.943** | **1.546** | **-0.090** | **0.066** | **0.18** |
| **Croston** | 1.168 | 1.738 | -0.365 | 0.267 | 7.65 |
| **SBA** | 1.138 | 1.699 | -0.279 | 0.204 | 6.83 |
| **TSB** | 0.981 | 1.548 | -0.208 | 0.152 | 7.08 |

Table X. Overall Performance Comparison of Forecasting Models

### Comparative Analysis

The comparative evaluation demonstrates clear differences in forecasting performance across the four forecasting methods. Among all evaluated models, the **4-week Moving Average (MA(4))** achieved the lowest forecasting error while also requiring the least computational time. Its combination of low MAE, lowest RMSE, relatively small forecasting bias, and minimal execution time establishes it as the strongest overall performer for the retail demand dataset.

Although the **TSB** model achieved forecasting accuracy comparable to the Moving Average baseline, the improvement was not sufficient to offset its substantially higher computational cost. Conversely, both **Croston's Method** and **SBA** produced higher forecasting errors and greater forecasting bias, indicating that their specialised intermittent-demand formulations did not provide additional forecasting benefit for this dataset.

### Model Selection

Based on the comparative evaluation, the **4-week Moving Average (MA(4))** was selected as the final forecasting model. The selection was based on four primary considerations:

- **Highest forecasting accuracy**, reflected by the lowest RMSE and MAE.

- **Lowest computational cost**, enabling efficient large-scale forecasting.
- **Lower forecasting bias** compared with specialised intermittent-demand methods.
- **Simple and interpretable implementation**, facilitating deployment and future maintenance.

Although more sophisticated intermittent-demand forecasting methods were evaluated, the experimental results demonstrate that increased model complexity did not produce meaningful improvements in forecasting performance for the demand patterns observed in this study.

### Key Takeaways

The comparative evaluation highlights the importance of validating forecasting models using empirical performance rather than assuming that specialised algorithms will always outperform simpler alternatives. Despite the intermittent nature of the retail demand data, the MA(4) model consistently achieved the best balance between forecasting accuracy, computational efficiency, and practical applicability. This finding reinforces the value of systematic model evaluation and demonstrates that the most suitable forecasting model is ultimately determined by the characteristics of the dataset rather than the complexity of the forecasting algorithm.

### Transition

Having identified the most suitable forecasting model, the following chapter summarises the overall findings of the study, discusses the business implications of the forecasting results, outlines the limitations of the current approach, and presents recommendations for future work.

# 8. Conclusion

## 8.1 Project Summary

### Overview

The objective of this project was to develop and evaluate forecasting methods capable of predicting weekly retail demand for individual store–product combinations using historical sales transactions. Beginning with daily transactional data, the project established a complete forecasting pipeline comprising data preparation, exploratory data analysis, feature engineering, forecasting model implementation, and comparative model evaluation. Each stage was designed to support reliable short-term demand forecasting while ensuring methodological consistency and practical applicability.

### Project Outcomes

The exploratory analysis identified several characteristics that significantly influenced forecasting behaviour, including intermittent demand patterns, promotional uplift, seasonal demand variation, and differences in demand across stores and products. These insights informed the selection of the forecasting granularity, the construction of the modelling dataset, and the evaluation methodology adopted throughout the study.

Four forecasting approaches were subsequently implemented and evaluated using a rolling-origin forecasting framework: the **4-week Moving Average (MA(4))**, **Croston's Method**, **Syntetos–Boylan Approximation (SBA)**, and the **Teunter–Syntetos–Babai (TSB)** method. Model performance was assessed using consistent accuracy, bias, and computational performance metrics to enable an objective comparison.

The comparative evaluation demonstrated that the **4-week Moving Average (MA(4))** achieved the strongest overall performance, providing the lowest forecasting error while maintaining minimal computational cost. Although specialised intermittent-demand forecasting methods were also evaluated, increased model complexity did not produce meaningful improvements for the retail demand characteristics observed in this dataset. Consequently, MA(4) was selected as the most suitable forecasting approach for the forecasting task considered in this study.

## 8.2 Key Findings

### Overview

The forecasting study produced several important findings regarding retail demand behaviour and forecasting model performance. These findings were derived from the exploratory data analysis and comparative evaluation of multiple forecasting methods and collectively explain the factors that influenced forecasting accuracy throughout the project.

### Key Findings

- **Retail demand exhibited strong intermittent characteristics.** A substantial proportion of store–product demand series contained frequent zero-demand periods interspersed with irregular purchasing events. This confirmed that conventional forecasting approaches designed for continuously demanded products may not always be appropriate and justified the evaluation of specialised intermittent-demand forecasting methods.
- **Promotional activity was the primary driver of short-term demand variation.** Products experiencing promotional price reductions consistently demonstrated higher average sales volumes than during non-promotional periods. However, the magnitude of promotional uplift varied considerably across stores and products, indicating that promotional effectiveness is highly context-dependent rather than uniform across the retail network.

- **Demand behaviour varied significantly across stores and products.** The exploratory analysis revealed considerable variability in purchasing behaviour between individual store–product combinations. This supported the decision to perform forecasting at the store–product–week level rather than using aggregated demand, thereby preserving local demand patterns and improving forecast relevance.
- **Specialised forecasting methods did not necessarily outperform simpler approaches.** Although Croston's Method, SBA, and TSB were specifically developed for intermittent demand forecasting, the comparative evaluation demonstrated that the **4-week Moving Average (MA(4))** achieved the strongest overall balance between forecasting accuracy, forecasting bias, and computational efficiency for the dataset considered in this study.
- **Forecasting methodology had a greater influence on model reliability than model complexity.**
  The use of a consistent rolling-origin evaluation framework ensured that every forecasting model was assessed under realistic operational conditions. This methodological consistency provided confidence that the observed performance differences reflected the forecasting methods themselves rather than variations in the evaluation procedure.

### Summary

Overall, the study demonstrates that successful retail demand forecasting depends on understanding the underlying demand characteristics, selecting an appropriate forecasting granularity, and evaluating models using a realistic forecasting framework. The findings also reinforce that empirical performance should guide model selection, as simpler forecasting methods may outperform more sophisticated alternatives when they better align with the characteristics of the available data.

## 8.3 Business Implications

### Overview

Accurate demand forecasting plays a critical role in retail operations by enabling better inventory planning, reducing operational costs, and improving product availability. The findings of this study demonstrate that selecting an appropriate forecasting methodology can provide practical benefits beyond forecasting accuracy alone. By identifying a forecasting model that is both accurate and computationally efficient, the proposed approach can support more effective decision-making across multiple retail functions.

### Operational Benefits

The selected **4-week Moving Average (MA(4))** model provides a simple and reliable approach for forecasting weekly demand at the store–product level. Its low computational requirements make it suitable for large-scale implementation across thousands of product-store combinations without requiring specialised forecasting infrastructure. This enables forecasts to be generated frequently while maintaining efficient resource utilisation.

Improved forecasting accuracy also supports inventory management by reducing the likelihood of both stock shortages and excess inventory. More reliable demand estimates allow retailers to align replenishment decisions more closely with expected customer demand, thereby improving product availability while reducing inventory holding costs.

### Business Value

The exploratory analysis further demonstrated that promotional activities and seasonal events have a measurable influence on demand behaviour. These insights enable retailers to anticipate periods of increased demand and allocate inventory more effectively during promotional campaigns and holiday periods. In addition, forecasting at the store–product–week level preserves local demand characteristics, allowing replenishment decisions to reflect differences in customer purchasing behaviour across individual retail locations.

### Strategic Perspective

Beyond identifying the most suitable forecasting model, the project establishes a repeatable forecasting framework that can support future analytical initiatives. The methodology developed during this study provides a foundation for evaluating additional forecasting models, integrating new data sources, and continuously improving forecasting performance as more historical demand data become available. As a result, the project contributes not only an effective forecasting model but also a structured forecasting process that can be extended to meet future business requirements.

## 8.4 Limitations

### Overview

While the forecasting framework developed in this study produced reliable results for the available retail demand data, several limitations should be considered when interpreting the findings. These limitations primarily arise from the characteristics of the available dataset and the scope of the forecasting methods evaluated, rather than deficiencies in the forecasting methodology itself.

### Data Availability

The forecasting models were developed using approximately seventy weeks of historical sales data. Although this period was sufficient for evaluating short-term forecasting performance, a longer historical time horizon would enable more comprehensive analysis of long-term seasonal behaviour, recurring promotional cycles, and year-over-year demand trends. Additional historical observations may therefore improve forecasting reliability for products exhibiting extended seasonal patterns.

### Limited Explanatory Variables

The modelling dataset primarily incorporated historical demand, pricing information, promotional indicators, and holiday-related variables. Other external factors that may influence retail demand, such as weather conditions, competitor pricing, marketing campaigns, regional events, or macroeconomic conditions, were not available within the scope of the project. Incorporating such variables may improve forecasting performance for products whose demand is strongly influenced by external factors.

### Forecasting Scope

The study focused on evaluating classical statistical forecasting methods suitable for intermittent demand. More advanced machine learning and deep learning approaches, such as gradient boosting or recurrent neural networks, were outside the scope of this project. Consequently, the findings should be interpreted as a comparison among the selected statistical forecasting methods rather than an exhaustive evaluation of all possible forecasting approaches.

### Model Generalisability

The conclusions presented in this study are specific to the anonymised retail dataset used during the internship. Although the forecasting methodology is transferable to similar retail environments, the relative performance of individual forecasting models may differ when applied to datasets with different demand characteristics, product portfolios, or promotional strategies. Model selection should therefore be validated whenever the methodology is deployed in a new business context.

## 8.5 Future Work

### Overview

The forecasting framework developed in this study provides a strong foundation for short-term retail demand forecasting. While the selected forecasting model achieved reliable performance for the available dataset, several opportunities exist to further enhance forecasting accuracy, scalability, and business value. Future work may focus on incorporating additional data sources, evaluating more advanced forecasting techniques, and extending the forecasting pipeline to support operational decision-making.

### Integration of Additional Data Sources

Future forecasting models could incorporate external variables that influence retail demand but were not available within the scope of this project. Examples include weather conditions, local events, competitor pricing, marketing campaigns, and macroeconomic indicators. Integrating these factors may improve forecast accuracy, particularly for products whose demand is strongly affected by external business conditions.

### Automated Forecasting Pipeline

The methodology developed during this project could be extended into an automated forecasting pipeline capable of generating forecasts on a regular schedule using newly available sales data. Such a system could integrate data preparation, model execution, performance monitoring, and forecast reporting into a single workflow, enabling continuous demand forecasting with minimal manual intervention.

### Operational Deployment

A practical extension of this work would involve integrating the forecasting framework into retail inventory planning and replenishment processes. Forecast outputs could support procurement decisions, inventory allocation, promotional planning, and stock replenishment at the store–product level. Continuous monitoring of forecasting performance would also allow periodic model retraining and evaluation as demand patterns evolve over time.

## 8.6 Key Learnings

### Overview

Beyond developing a retail demand forecasting solution, this project provided valuable insights into the practical aspects of building reliable forecasting systems. The experience extended beyond implementing forecasting algorithms and emphasised the importance of understanding business problems, validating methodologies, interpreting data, and making evidence-based modelling decisions. These lessons will serve as a strong foundation for future work in data science and forecasting.

### Importance of Methodological Rigor

One of the most significant learnings from this project was the importance of validating the forecasting methodology before interpreting model performance. Reliable forecasting depends not only on selecting an appropriate algorithm but also on ensuring that the evaluation process accurately reflects real-world forecasting conditions. Implementing a rolling-origin evaluation framework reinforced the value of preventing information leakage and highlighted that methodological correctness is essential for producing trustworthy forecasting results.

### Value of Data Understanding

The exploratory data analysis demonstrated that understanding the characteristics of the data is fundamental to successful forecasting. Analysing demand intermittency, promotional behaviour, seasonal effects, and variability across stores and products provided the context required to interpret forecasting performance and justify modelling decisions. This project reinforced that effective forecasting begins with understanding the data rather than immediately applying complex algorithms.

## Balancing Simplicity and Performance

An important outcome of the comparative evaluation was the recognition that increased model complexity does not necessarily lead to superior forecasting performance. Despite evaluating specialised intermittent-demand forecasting methods, the four-week Moving Average model achieved the strongest overall balance between forecasting accuracy, computational efficiency, and implementation simplicity. This emphasised the importance of selecting models based on empirical evidence rather than perceived sophistication.

## Professional Development

This project strengthened practical skills in data preparation, exploratory data analysis, time-series forecasting, model evaluation, and technical documentation. Equally important, it reinforced the value of iterative development, critical evaluation, and continuous improvement throughout the analytical process. The experience demonstrated that successful data science projects require not only technical implementation but also clear communication, reproducible methodologies, and well-justified analytical decisions.

## Closing Reflection

The knowledge and experience gained throughout this project extend beyond the implementation of a forecasting model. Developing an end-to-end forecasting pipeline—from raw transactional data to model evaluation and technical documentation—provided practical insight into how data science methodologies are applied to solve real-world business problems. The lessons learned during this project will contribute to future work in forecasting, analytics, and data-driven decision-making.